

Marcus Lai

CONTACT INFORMATION

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marclai2003.github.io

EDUCATION

University of British Columbia, Vancouver, BC

BSc in *Honours Computer Science*

Sept 2021 - May 2026

- **GPA:** 3.88/4, **cGPA** 89.9%.
- **Science Scholar (Year Average ≥ 90):** 2024
- **Dean's List (Year Average ≥ 80):** 2021, 2022, 2023
- **Thesis:** Approximation Algorithms in Multicommodity Network Flow with Prof. Bruce Shepherd

RESEARCH EXPERIENCE

Error Detection in the Linux Kernel with eBPF

Prof. Thomas Pasquier @ UBC Systopia Lab

May 2025 - Sept 2025

- Enhanced detection coverage of kernel errors in the Linux Kernel with eBPF, and improved existing detection templates in **C**.
- Designed and implemented automation steps in the pipeline from Syzkaller, a Linux kernel fuzzer, to detector eBPF programs.

Optimal Cost-sensitive Decision Trees

Prof. Margo Seltzer @ UBC Systopia Lab

Sept 2024 - May 2025

- Extended the UBC Systopia GOSDT algorithm with a novel objective function that incorporates feature costs.
- Reworked proofs for 10 pruning bounds to adapt to the new objective function, demonstrating theoretical correctness.
- Derived new pruning bounds unique to the feature-cost setting for further pruning.
- Implemented the new objective function on top of the GOSDT repository in **C++**.

Bi-level Decision Tree Heuristic (in submission)

PhD student Chenxuanyin Zou @ Cao Research Group

March 2024

ICLR 2026 #8113

- Implemented an acceleration of the CART algorithm in **Julia** by linear and binary threshold pruning, achieving 10% training speedup across 64 datasets.

Cache Sidechannel Attacks on Serverless Applications

Prof. Aastha Mehta @ UBC Systopia Lab

May 2024 - Sept 2024

- Designed and performed cache latency and multicore synchronization experiments to reverse-engineer undocumented behaviors in Intel's Icelake architecture, such as the existence of a L2-remote cache access and cache directory data structure.
- Implemented flushed+reload and prime+probe-style cache attacks on a non-inclusive cache architecture in lab servers.

Modelling Chirality in 4-cell stage of Roundworm *C. Elegans* (in submission)

Prof. Eric Cybrynbaum @ Cytryn Lab

May 2023 - Sept 2023

Developmental Cell *EMID#713d1c5a8836c1ed*

- Developed a physics-based ODE model simulating embryonic development of *C. elegans* at the 4-cell stage to test hypotheses related to chirality.

- Implemented a **Python** simulation suite, with SciPy for parameter fitting and Matplotlib animations for visualization of model-dynamics.

Generalizations of Artin’s Constant (PIMS VXML)

Prof. Greg Martin

Sept 2022 - Dec 2022

- Analytically derived the density of primes from the Goldmakher-Martin conjecture. Calculated these constants to high precision and implemented techniques to approximate the error on truncated sums and products.
- Compared empirical data against analytical results to reinforce the conjecture.
- My observations of a recursive pattern in the derivatives led to a closed-form formula to calculate the conjecture constants, which is incorporated into future works.

ADDITIONAL EXPERIENCE

Undergraduate Teaching Assistance (UTA)

Math 100 (Calculus I), 101 (Calculus II), 110 (Calculus I)

Sept 2022 - May 2025

- Delivered supplementary lecture content in tutorials to reinforce students’ understanding of core calculus concepts.
- Led group discussions and guided problem-solving sessions to promote active learning and peer collaboration.
- Assisted with assignment grading by providing constructive, detailed feedback.

AWARDS

Undergraduate Student Research Award (NSERC USRA)

Summer research award valued at 10,000.

May 2023, 2024
2025

PROGRAMMING LANGUAGES

Proficient: Python, C++, C, Julia, JavaScript / Typescript, HTML, CSS

Familiar: Java, MATLAB